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Filter device

Abstract

An object of this invention is to reduce the size of a filter device. Provided is a filter device (**1**) including: a substrate (**11**) including a first main surface and a second main surface (**111, 112**); strip-shaped conductors (**12***a***1** to **12***a***5**) provided to the first main surface (**111**); and a ground conductor layer (**13**) provided at least to the second main surface (**112**), the second main surface (**112**) having a recessed portion (**11***ai*) provided for each strip-shaped conductor (**12***ai*), the recessed portion (**11***ai*) overlapping the strip-shaped conductor (**12***ai*) in plan view and having a surface covered with the ground conductor layer (**13**).

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Background/Summary

TECHNICAL FIELD
(1) The present invention relates to a filter device.
BACKGROUND ART
(2) FIGS. 3 and 4 of Patent Literature 1 disclose, as a conventional example, a microstrip filter device (in Patent Literature 1, a resonant circuit device) including: a substrate made of a dielectric (in Patent Literature 1, a dielectric substrate 1); strip-shaped conductors which are provided to a first main surface of the substrate and adjacent ones of which are electromagnetically coupled to each other (in Patent Literature 1, resonant conductors 3 to 7); and a ground conductor layer provided to a second main surface of the substrate (in Patent Literature 1, a ground conductor 2). Note that each of the strip-shaped conductors functions as a resonator.
CITATION LIST
Patent Literature
Patent Literature 1
(3) Japanese Patent Application Publication, Tokukaihei, No. 9-139605 (1997)
SUMMARY OF INVENTION

Technical Problem
(4) In addition, FIG. 1 of Patent Literature 1 illustrates the filter device including recessed portions (in FIG. 1 of Patent Literature 1, trenches 11) that are provided in areas of the substrate which areas do not overlap the strip-shaped conductors (in FIG. 1 of Patent Literature 1, resonant conductors 5 and 6) in plan view and that are opened in the first main surface. With this configuration, since a specific inductive capacity of air filled in the recessed portions is smaller than a specific inductive capacity of the dielectric constituting the substrate, it is possible to reduce a degree of electromagnetic coupling between adjacent ones of the strip-shaped conductors. Thus, if the filter device is designed such that the degree of coupling between the adjacent ones of the strip-shaped conductors is substantially the same as those in conventional ones, a distance between the adjacent ones of the strip-shaped conductors can be reduced. Therefore, the filter device can be reduced in size. Such a filter device, however, is required to be further reduced in size.
(5) An aspect of the present invention was made in consideration of the above-described problem, and has an object to reduce a filter device in size as compared to conventional ones.
Solution to Problem
(6) A filter device in accordance with a first aspect of the present invention includes: a substrate which is made of a dielectric and which includes a first main surface and a second main surface facing each other; strip-shaped conductors which are provided to the first main surface and adjacent ones of which are electromagnetically coupled to each other; and a ground conductor layer provided at least to the second main surface, wherein in the second main surface of the substrate, one or more recessed portions are provided for each of the strip-shaped conductors, the one or more recessed portions overlapping the each of the strip-shaped conductors when seen in plan view, the one or more recessed portions having a surface covered with the ground conductor layer.
Advantageous Effects of Invention
(7) A filter device in accordance with an aspect of the present invention can be reduced in size.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) (a) of FIG. **1** is a plan view of a filter device in accordance with Embodiment 1 of the present invention. (b) and (c) of FIG. **1** are cross-section views of the filter device shown in (a) of FIG. **1**.
(2) FIG. **2** is a cross-section view of Variation 1 of the filter device shown in FIG. **1**.
(3) (a) and (b) of FIG. **3** are respectively a plan view and a cross-section view of Variation 2 of the filter device shown in FIG. **1**.
(4) FIG. **4** is a cross-section view of Variation 3 of the filter device shown in FIG. **1**.
(5) FIG. **5** is a cross-section view of Variation 4 of the filter device shown in FIG. **1**.
(6) (a) of FIG. **6** is a plan view of a filter device in accordance with Embodiment 2 of the present invention. (b) and (c) of FIG. **6** are cross-section views of the filter device shown in (a) of FIG. **6**.
(7) FIG. **7** is an enlarged plan view of one end portion of a strip-shaped conductor included in a variation of the filter device shown in FIG. **6**.
(8) FIG. **8** is a plan view of a filter device in accordance with an Example of the present invention.
(9) FIG. **9** is a graph illustrating frequency dependencies of transmission intensities of filter devices in accordance with the Example of the present invention, Comparative Example 1, and Comparative Example 2.
DESCRIPTION OF EMBODIMENTS
(10) A filter device in accordance with an embodiment of the present invention functions as a bandpass filter that allows passage of, of high frequency signals having a frequency within a frequency band which is called a millimeter wave or a microwave, a high frequency signal within a given pass band and that blocks the other high frequency signals. The description of the later-

described Embodiment 1 and Embodiment 2 will be given based on an assumption that a center frequency of the pass band is included in a 25-GHz band. However, the center frequency and the bandwidth of the pass band are not limited to any particular ones, and may be designed as appropriate according to the purpose of use of the filter device.

Embodiment 1

(11) The following description will discuss, with reference to FIG. **1**, a filter device in accordance with Embodiment 1 of the present invention. (a) of FIG. **1** is a plan view of the filter device **1**. (b) and (c) of FIG. **1** are cross-section views of the filter device **1**. (b) of FIG. **1** is a cross-section view illustrating a cross section taken along A-A′ line shown in (a) of FIG. **1**, and (c) of FIG. **1** is a cross-section view illustrating a cross section taken along B-B′ line shown in (a) of FIG. **1**.

(12) <Configuration of Filter Device>

(13) As shown in (a) to (c) of FIG. **1**, the filter device **1** includes a substrate **11**, a conductor pattern **12**, and a ground conductor layer **13**.

(14) (Substrate)

(15) The substrate **11** is a plate-like member which is made of a dielectric and which includes a main surface **111** and a main surface **112** facing each other. The main surface **111** is an example of the first main surface recited in the claims. The main surface **112** is an example of the second main surface recited in the claims.

(16) In Embodiment 1, the substrate **11** is made of quartz. However, the dielectric constituting the substrate **11** is not limited to quartz, but can be selected as appropriate. Examples of the dielectric encompass glass other than quartz, ceramic, semiconductors such as silicon and GaAs, and resins.

(17) In Embodiment 1, the substrate **11** has a rectangular shape when the main surface **111** is seen in plan view along a line normal to the main surface **111**. However, the shape of the substrate **11** is not limited to the rectangular shape, but can be selected as appropriate. In the description below, the expression “seeing in plan view” refers to seeing the main surface **111** along a line normal to the main surface **111**.

(18) In Embodiment 1, the main surface **111** is provided with the later-described conductor pattern **12**, and the main surface **112** is provided with the recessed portions **11*a*1** to **11*a*5** and the ground conductor layer **13** (described later). Alternatively, the conductor pattern **12** may be indirectly provided to the main surface **111** of the substrate **11**, and the ground conductor layer **13** may be indirectly provided to the main surface **112** of the substrate **11**. For example, another layer having a low conductivity (e.g., a dielectric layer) may be interposed (i) between the main surface **111** and the conductor pattern **12** and/or (ii) between the main surface **112** and the ground conductor layer **13**. The substrate **11** includes, in its inside, the later-described conductor posts **11*b*1** to **11*b*5**.

(19) (Conductor Pattern)

(20) The conductor pattern **12** provided to the main surface **111** can be obtained by patterning of a conductor film into a given shape. In Embodiment 1, the conductor pattern **12** is made of copper. However, the conductor constituting the conductor pattern **12** is not limited to copper, but can be selected as appropriate. The conductor pattern **12** includes strip-shaped conductors **12*a*1** to **12*a*5**, a coplanar line **12*b***, and a coplanar line **12*c***. In Embodiment 1, the conductor pattern **12** is constituted by five strip-shaped conductors **12*a*1** to **12*a*5**. However, the number of strip-shaped conductors constituting the conductor pattern **12** is not limited to five.

(21) As shown in (a) of FIG. **1**, each of the strip-shaped conductors **12*a*1** to **12*a*5** has a rectangular shape. Hereinafter, a direction in which the strip-shaped conductors **12*a*1** (i is an integer of not less than one and not more than five) extend (i.e., a direction extending along longer sides of the strip-shaped conductors **12*a*1**) will be referred to as a lengthwise direction. Meanwhile, a direction crossing the lengthwise direction (i.e., a direction extending along shorter sides of the strip-shaped conductors **12*a*1**) will be referred to as a width direction. In each strip-shaped conductor **12*a*1**, a length measured in the lengthwise direction will be referred to as a length, whereas a length measured in the width direction will be referred to as a width.

(22) The strip-shaped conductors **12*a*1** are arranged such that their longer sides are in parallel with each other. Further, the strip-shaped conductors **12*a*1** are arranged such that a distance between adjacent ones of the strip-shaped conductors has a certain value. Each of the strip-shaped conductors **12*a*1** arranged in this manner is electromagnetically coupled to another one of the strip-shaped conductors **12*a*1** adjacent to the each of the strip-shaped conductors **12*a*1**. A distance between adjacent ones of the strip-shaped conductors is adjusted as appropriate so that a degree of coupling between adjacent ones of the strip-shaped conductors attains a desired value.
(23) When seen along the lengthwise direction of each strip-shaped conductor **12*a*1**, a length of each strip-shaped conductor **12*a*1** can be defined as appropriate in accordance with a center frequency of a pass band and a specific inductive capacity of the substrate **11**. In Embodiment 1, the length of each strip-shaped conductor **12*a*1** is defined to be a quarter of an effective wavelength of an electromagnetic wave whose frequency is equal to the center frequency. However, the length of each strip-shaped conductor **12*a*1** is not limited to the quarter of the effective wavelength, and may alternatively be an integral multiple of the quarter.
(24) The coplanar line **12*b*** is made of a signal line **12*b*1** and ground conductor patterns **12*b*2** and **12*b*3**. One end portion of the signal line **12*b*1** is electrically connected to one end portion of the strip-shaped conductor **12*a*1**. The ground conductor patterns **12*b*2** and **12*b*3** are disposed such that the signal line **12*b*1** is sandwiched therebetween. The coplanar line **12*b*** functions as an input-output port of the filter device **1**.
(25) The coplanar line **12*c*** is made of a signal line **12*c*1** and ground conductor patterns **12*c*2** and **12*c*3**. One end portion of the signal line **12*c*1** is electrically connected to one end portion of the strip-shaped conductor **12*a*5**. The ground conductor patterns **12*c*2** and **12*c*3** are disposed such that the signal line **12*c*1** is sandwiched therebetween. The coplanar line **12*c*** functions as an input-output port of the filter device **1**.
(26) (Recessed Portion)
(27) The recessed portions **11*a*1** to **11*a*5** provided to the main surface **112** respectively correspond to the strip-shaped conductors **12*a*1** to **12*a*5** facing thereto. Each recessed portion **11*ai*** corresponding to its respective strip-shaped conductor **12*a*1** is disposed so as to overlap its respective strip-shaped conductor **12*a*1** when the main surface **111** is seen in plan view (see (a) of FIG. **1**). In Embodiment 1, each recessed portion **11*ai*** is disposed so as to cover its respective strip-shaped conductor **12*a*1**. Note that each recessed portion **11*ai*** only needs to at least partially overlap at least a part of its respective strip-shaped conductor **12*ai***.
(28) Each recessed portion **11*ai*** has a bottom surface and a side surface, which constitute a surface of the each recessed portion **11*ai*** and which are covered with the later-described second ground conductor layer **132** (see (b) of FIG. **1**).
(29) In Embodiment 1, each recessed portion **11*ai*** has a rectangular parallelepiped shape. However, the shape of each recessed portion **11*ai*** is not limited to the rectangular parallelepiped, and can be selected as appropriate.
(30) In Embodiment 1, a width of each recessed portion **11*ai*** is greater than a width of a strip-shaped conductor **12*ai*** corresponding to the recessed portion **11*ai***. Alternatively, the width of each recessed portion **11*ai*** may be either smaller than or equal to the width of its respective strip-shaped conductor **12*ai***.
(31) Note that a distance between each strip-shaped conductor **12*a*1** and the bottom surface of its respective recessed portion **11*ai*** is adjusted as appropriate to give a desired degree of coupling between the each strip-shaped conductor **12*a*1** and a portion of the second ground conductor layer **132** which portion is provided to the bottom surface of its respective recessed portion **11*ai***.
(32) In Embodiment 1, when seen along the lengthwise direction in which each strip-shaped conductor **12*ai*** extends, a length of each recessed portion **11*ai*** is longer than a length of its respective strip-shaped conductor **12*ai*** overlapping, in plan view, the each recessed portion **11*ai*** (see (c) of FIG. **1**). When seen along the lengthwise direction of each strip-shaped conductor **12*a*1**,

each recessed portion **11***ai* covers its respective strip-shaped conductor **12***ai* overlapping the each recessed portion **11***ai* (see (a) of FIG. **1**).
(33) (Ground Conductor Layer)
(34) The ground conductor layer **13** is provided at least on the main surface **112**. To be more specific, as shown in (b) of FIG. **1**, the ground conductor layer **13** is constituted by a first ground conductor layer **131** and a second ground conductor layer **132**. The first ground conductor layer **131** refers to a portion of the ground conductor layer **13** which portion is provided to the main surface **112**, whereas the second ground conductor layer **132** refers to a portion of the ground conductor layer **13** which portion covers a surface of each recessed portion **11***ai*.
(35) The ground conductor layer **13** is made of a conductor film. In Embodiment 1, the ground conductor layer **13** is made of copper. However, the conductor constituting the ground conductor layer **13** is not limited to copper, but can be selected as appropriate.
(36) As shown in (b) of FIG. **1**, the first ground conductor layer **131** and the second ground conductor layer **132** are formed so as to be continuous to each other, and are electrically connected to each other. Thus, a potential of the first ground conductor layer **131** and a potential of the second ground conductor layer **132** are identical to each other.
(37) (Conductor Post)
(38) The conductor posts **11***b***1** to **11***b***5** respectively correspond to the strip-shaped conductors **12***a***1** to **12***a***5**. Each conductor post **11***bi* corresponding to its respective strip-shaped conductor **12***a***1** is disposed in an area (in Embodiment 1, the one end portion) in which its respective strip-shaped conductor **12***a***1** and its respective recessed portion **11***ai* overlap each other when the main surface **111** is seen in plan view (see (a) of FIG. **1**), and short-circuits the respective strip-shaped conductor **12***a***1** and the second ground conductor layer **132** (see the conductor posts **11***b***2** and **11***b***4** shown in (b) of FIG. **1**).
(39) Each conductor post **11***bi* can be obtained by forming a conductor film on an inner wall of a through-hole provided in an area of the substrate **11** which area corresponds to the one end portion of its respective strip-shaped conductor **12***a***1**. Alternatively, each conductor post **11***bi* may be made of a conductor filled in the through-hole.
(40) In Embodiment 1, when the main surface **111** is seen in plan view, each recessed portion **11***ai* covers it respective strip-shaped conductor **12***a***1**. Thus, each conductor post **11***bi* is located inside its respective recessed portion **11***ai* in plan view. However, the position where each conductor post **11***bi* is provided is not limited to the position inside its respective recessed portion **11***ai*, and may alternatively be a position outside its respective recessed portion **11***ai* (i.e., the first ground conductor layer) or a position on an outer periphery (i.e., a side surface) of its respective recessed portion **11***ai*.
(41) When seen in plan view, conductor posts **11***c***1**, **11***c***2**, **11***c***3**, and **11***c***4** are respectively disposed in areas overlapping the ground conductor patterns **12***b***2**, **12***b***3**, **12***c***2**, and **12***c***3**. The conductor posts **11***c***1**, **11***c***2**, **11***c***3**, and **11***c***4** respectively short-circuit the ground conductor patterns **12***b***2**, **12***b***3**, **12***c***2**, and **12***c***3** to the first ground conductor layer **131**.
(42) Note that, in the filter device **1**, each conductor post **11***bi* is constituted by two conductor posts. However, there is no limitation on the number of conductor posts constituting each conductor post **11***bi*, and the number of conductor posts constituting each conductor post **11***bi* may be one or three or more. A cross-sectional shape of the conductor post(s) constituting each conductor post **11***bi* is not limited to a circle.
(43) <Variation 1>
(44) Next, the following description will discuss, with reference to FIG. **2**, a filter device **1**A, which is Variation 1 of the filter device **1** shown in FIG. **1**. FIG. **2** is a cross-section view of the filter device **1**A, and this cross-section view corresponds to the cross-section view of the filter device **1** shown in (b) of FIG. **1**. Note that, for convenience, members of the filter device **1**A having functions identical to those of the respective members described for the filter device **1** are given

respective identical reference numerals, and a description of those members is omitted here. This also applies to the later-described variations.
(45) The filter device **1**A can be obtained by modifying the filter device **1** such that the shape of each recessed portion **11***ai* is changed from a rectangular parallelepiped shape to a half-pipe shape. The “half-pipe shape” herein refers to a shape obtained by cutting, along a center axis of the pipe and along a shorter axis of the ellipse, a pipe having an ellipse cross section into two.
(46) In the filter device **1**, each recessed portion **11***ai* has a rectangular parallelepiped shape. Therefore, a non-continuous angle is formed at a boundary between the bottom surface and the side surface of each recessed portion **11***ai* (see (b) of FIG. **1**). Alternatively, the bottom surface and the side surface of each recessed portion **11***ai* may be smoothly connected to each other, as in the filter device **1**A in accordance with Variation 1. Further alternatively, another variation of the filter device **1**A can be obtained by modifying the filter device **1** such that each recessed portion **11***ai*, which has a rectangular parallelepiped shape, is transformed into a shape having a circular-arc (e.g., a half-circle) bottom surface.
(47) Variation 1 can also bring about similar effects given by Embodiment 1. Furthermore, in a case where the recessed portions **11***ai* having a half-pipe shape are employed as in Variation 1, it is possible to bring about another effect of facilitating forming of a second ground conductor layer **132** having a uniform thickness in the recessed portions **11***ai*, as compared to a case where the recessed portions **11***ai* having a rectangular parallelepiped shape are employed.
(48) <Variation 2>
(49) Next, the following description will discuss, with reference to FIG. **3**, a filter device **1**B, which is Variation 2 of the filter device **1** shown in FIG. **1**. (a) of FIG. **3** is a plan view of the filter device **1**B. (b) of FIG. **3** is an A-A′ cross-section view of the filter device **1**B, and this cross-section view corresponds to the cross-section view of the filter device **1** shown in (b) of FIG. **1**.
(50) The filter device **1**B can be obtained by modifying the filter device **1** such that the shape of each recessed portion **11***ai* is changed from a rectangular parallelepiped shape to an E-shape when the main surface **111** is seen in plan view. Thus, the description in Variation 2 will discuss the shape of each recessed portion **11***ai*.
(51) As shown in FIG. **3**, each recessed portion **11***ai* of the filter device **1**B is constituted by a first recessed portion **11***ai***1**, a second recessed portion **11***ai***2**, a third recessed portion **11***ai***3**, and a fourth recessed portion **11***ai***4**.
(52) Each of the first recessed portion **11***ai***1**, the second recessed portion **11***ai***2**, and the third recessed portion **11***ai***3** has a rectangular parallelepiped shape, similarly to each recessed portion **11***ai* in the filter device **1**. Note that each of the first recessed portion **11***ai***1**, the second recessed portion **11***ai***2**, and the third recessed portion **11***ai***3** has a width that is an approximately one-fifth of the width of each recessed portion **11***ai* in the filter device **1**. Further, the first recessed portion **11***ai***1**, the second recessed portion **11***ai***2**, and the third recessed portion **11***ai***3** are disposed at equal intervals. Note that the width of each recessed portion **11***ai* in the filter device **1**B is equal to the width of each recessed portion ai in the filter device **1**.
(53) The fourth recessed portion **11***ai***4** is disposed in an area including conductor posts **11***bi* when the main surface **111** is seen in plan view. The fourth recessed portion **11***ai***4** is disposed such that its longitudinal direction extends along a width direction of its respective strip-shaped conductor **12***a***1** so as to allow the first recessed portion **11***ai***1**, the second recessed portion **11***ai***2**, and the third recessed portion **11***ai***3** to communicate with each other via the fourth recessed portion **11***ai***4**. A distance between (i) a bottom surface of the fourth recessed portion **11***ai***4** which bottom surface is an area being included in the bottom surface of the recessed portion **11***ai* and including the conductor posts **11***bi* and (ii) the main surface **111** is constant.
(54) Note that the filter device **1**B may not include the fourth recessed portion **11***ai***4**. In this case, each recessed portion **11***ai* in the filter device **1**B is constituted by three recessed portions, i.e., the first recessed portion **11***ai***1**, the second recessed portion **11***ai***2**, and the third recessed portion **11***ai***3**.

(55) Variation 2 can also bring about similar effects given by Embodiment 1. Furthermore, in a case where the recessed portions **11***ai* each constituted by a plurality of divided recessed portions are employed as in Variation 2, it is possible to bring about another effect of facilitating manufacturing of recessed portions, as compared to a case in which the recessed portions each constituted by a single recessed portion are employed. In addition, providing a plurality of small-width recessed portions can bring about further another effect of enhancing the strength of the substrate **11**, as compared to the filter device **1**. Note that the feature of the foregoing Variation 1 can also be applied to Variation 2. With this, Variation 2 can additionally bring about the effects of Variation 1.
(56) <Variations 3 and 4>
(57) Next, the following description will discuss, with reference to FIG. **4**, a filter device **1**C, which is Variation 3 of the filter device **1** shown in FIG. **1**. In addition, the description of a filter device **1**D, which is Variation 4 of the filter device **1** shown in FIG. **1**, will also be given with reference to FIG. **5**. FIG. **4** is a cross-section view of the filter device **1**C, and this cross-section view corresponds to the cross-section view of the filter device **1** shown in (b) of FIG. **1**. FIG. **5** is a cross-section view of the filter device **1**D, and this cross-section view corresponds to the cross-section view of the filter device **1** shown in (b) of FIG. **1**.
(58) As shown in FIG. **4**, the filter device **1**C can be obtained by modifying the filter device **1** so as to further include a shield **14** made of a metal. The shield **14** can be obtained by forming (e.g., press forming) of a metal plate. The shield **14** has a top plate provided along a main surface **111** and a side wall provided to surround the sides of the top plate. The top plate covers strip-shaped conductors **12***a***1** while keeping a distance from the strip-shaped conductors **12***a***1**. The side wall surrounds the sides of the strip-shaped conductors **12***ai*.
(59) The main surface **111** of the substrate **11** constituting the filter device **1**C is provided with a strip-shaped conductor **12***d* surrounding an outer periphery of the main surface **111**. The strip-shaped conductor **12***d* is short-circuited to a first ground conductor layer **131** via a conductor post(s) not illustrated in FIG. **4**.
(60) A lower end of the side wall of the shield **14** is fixed to the strip-shaped conductor **12***d* by soldering (not illustrated in FIG. **4**), which is an example of a connecting member. Note that the connecting member only needs to be a member having electric conductivity and being capable of fixing metal pieces to each other, and is not limited to the soldering. Another example of the connecting member is silver paste. The shield **14** configured in this manner is short-circuited to the first ground conductor layer **131** via the strip-shaped conductor **12***d* and the conductor post(s).
(61) As shown in FIG. **5**, a filter device **1**D can be obtained by modifying the filter device **1** so as to further include a shield **15**. The shield **15** includes a substrate **15***a* made of a dielectric, a plurality of conductor posts **15***b* provided in an outer periphery of the substrate **15***a* arranged in a fence-like manner, and a conductor layer **15***c*.
(62) The conductor layer **15***c* is disposed so as to cover a main surface farther from the substrate **11** among a pair of main surfaces of the substrate **15***a*. The conductor layer **15***c* corresponds to the top plate of the shield **14**, and covers strip-shaped conductors **12***a***1** while keeping a distance from the strip-shaped conductors **12***ai*.
(63) Each of the conductor posts **15***b* can be obtained by forming a conductor film on an inner wall of a through-hole penetrating through the main surfaces of the substrate **15***a* or by filling a conductor in the through-hole. A center-to-center distance between adjacent ones of the conductor posts **15***b* can be defined as appropriate, and is preferably defined so as to be capable of reflecting an electromagnetic wave within a given pass band (e.g., 25 GHz band). The conductor posts arranged at intervals of such a center-to-center distance function as a post wall, and function similarly to the side wall of the shield **14**.
(64) Note that the filter device **1**D employs a bump **16** as a connecting member for fixing the conductor posts **15***b* to a strip-shaped conductor **12***d*. However, the connecting member is not limited to the bump **16**, and may alternatively be soldering or a solder ball.

Another Aspect of the Present Invention

(65) The description in Embodiment 1 has dealt with the filter device **1**, which is an aspect of the present invention. However, an aspect of the present invention is not limited to the filter device **1**. That is, the present invention encompasses the below-described features of the filter device **1**.

(66) A transmission line that is an aspect of the present invention is a transmission line constituting a part of the filter device **1** shown in FIG. **1**, and includes: a substrate **11** which is made of a dielectric and which includes a first main surface **111** and a second main surface **112** facing each other; a single strip-shaped conductor (here, a strip-shaped conductor **12*a*2**) provided to the first main surface **111**; and a ground conductor layer **13** provided at least on the second main surface **112**. In the transmission line, the second main surface **112** has one or more recessed portions (here, a recessed portion **11*a*2**) which overlap the strip-shaped conductor **12*a*2** in plan view and which have a surface covered with the second ground conductor layer **132** of the ground conductor layer **13**. The transmission line is a microstrip transmission line. In this configuration, a distance between the strip-shaped conductor and the ground conductor layer can be reduced, as compared to a microstrip transmission line including: a substrate not provided with one or more recessed portions; a single strip-shaped conductor; and a ground conductor layer. Thus, the strip-shaped conductor can be reduced in width, which makes it possible to reduce the transmission line in size in a width direction of the strip-shaped conductor. Further, in a case where a microstrip transmission line configured as above is employed as each of transmission lines in a configuration in which the microstrip transmission lines are arranged in parallel, a distance between strip-shaped conductors in adjacent ones of the transmission lines can be reduced.

(67) The transmission line functions as a resonator having a resonance frequency defined in accordance with the length of the strip-shaped conductor **12*a*2**. Thus, the present invention encompasses a resonator including: a substrate **11** which is made of a dielectric and which includes a first main surface **111** and a second main surface **112** facing each other; a single strip-shaped conductor (here, a strip-shaped conductor **12*a*2**) provided to the first main surface **111**; and a ground conductor layer **13** provided at least on the second main surface **112**, wherein the second main surface **112** has one or more recessed portions (here, a recessed portion **11*a*2**) which overlap the strip-shaped conductor **12*a*1** in plan view and which has a surface covered with a second ground conductor layer **132** of the ground conductor layer **13**. According to this configuration, a distance between the strip-shaped conductor and the ground conductor layer can be reduced, as compared to a microstrip resonator including: a substrate not provided with one or more recessed portions; a single strip-shaped conductor; and a ground conductor layer. Thus, the strip-shaped conductor can be reduced in width, which makes it possible to reduce the resonator in size in a width direction of the strip-shaped conductor. Further, in a case where a microstrip resonator configured as above is employed as each of microstrip resonators in a configuration in which the microstrip resonators are arranged substantially in parallel, a distance between strip-shaped conductors in adjacent ones of the resonators can be reduced.

(68) A transmission line group that is an aspect of the present invention is a transmission line group constituting a part of the filter device **1** shown in FIG. **1**, and includes: a substrate **11** which is made of a dielectric and which includes a first main surface **111** and a second main surface **112** facing each other; strip-shaped conductors (here, strip-shaped conductors **12*a*2** and **12*a*3**) which are provided to the first main surface **111** and which are adjacent to each other; and a ground conductor layer **13** provided at least on the second main surface **112**, wherein the second main surface **112** has one or more recessed portions (here, recessed portions **11*a*2** and **11*a*3**) which are provided for each of the strip-shaped conductors (here, the strip-shaped conductors **12*a*2** and **12*a*3**) so as to overlap the strip-shaped conductor **12*a*2** in plan view and which have a surface covered with a second ground conductor layer **132** of the ground conductor layer **13**.

(69) The transmission line group functions as a resonator group having a resonance frequency defined in accordance with lengths of the strip-shaped conductors **12*a*2** and **12*a*3**. Thus, the present

invention encompasses a resonator group including: a substrate **11** which is made of a dielectric and which includes a first main surface **111** and a second main surface **112** facing each other; strip-shaped conductors (here, strip-shaped conductors **12***a***2** and **12***a***3**) which are provided to the first main surface **111** and which are adjacent to each other; and a ground conductor layer **13** provided at least to the second main surface **112**, wherein the second main surface **112** has one or more recessed portions (here, recessed portions **11***a***2** and **11***a***3**) which are provided for each of the strip-shaped conductors (here, the strip-shaped conductors **12***a***2** and **12***a***3**), the one or more recessed portions overlapping the strip-shaped conductor **12***a***2** when seen in plan view and which have a surface covered with a second ground conductor layer **132** of the ground conductor layer **13**.

(70) An aspect of the present invention is not limited to the transmission line, the resonator, the transmission line group, or the resonator group each of which is limited to constitute a part of the filter device **1**. Alternatively, an aspect of the present invention may be any of a transmission line, a resonator, a transmission line group, and a resonator group each of which constitutes a part of any of the filter device **1**A, the filter device **1**B, the later-described filter device **2**, and the later-described filter device **2**A.

Embodiment 2

(71) The following description will discuss, with reference to FIG. **6**, a filter device **2** in accordance with Embodiment 2 of the present invention. (a) of FIG. **6** is a plan view of the filter device **2**. (b) and (c) of FIG. **6** are cross-section views of the filter device **2**. (b) of FIG. **6** is a cross-section view illustrating a cross section taken along A-A′ line shown in (a) of FIG. **6**, and (c) of FIG. **6** is a cross-section view illustrating a cross section taken along B-B′ line shown in (a) of FIG. **6**.

(72) <Configuration of Filter Device>

(73) As shown in (a) to (c) of FIG. **6**, the filter device **2** includes a substrate **21**, a conductor pattern **22**, and a ground conductor layer **23**. The substrate **21**, the conductor pattern **22**, and the ground conductor layer **23** of the filter device **2** respectively correspond to the substrate **11**, the conductor pattern **12**, and the ground conductor layer **13** of the filter device **1**. Therefore, the following description will discuss features of the filter device **2** that are different from those of the filter device **1**, and does not discuss features of the filter device **2** that are identical to those of the filter device **1**.

(74) (Substrate)

(75) Similarly to the substrate **11**, the substrate **21** is a plate-like member which is made of a dielectric and which includes a main surface **211** and a main surface **212** facing each other. The main surface **211** and the main surface **212** respectively correspond to the main surface **111** and the main surface **112** of the substrate **11**.

(76) In Embodiment 2, the main surface **211** is provided with the later-described conductor pattern **22**, and the main surface **212** is provided with the recessed portions **21***a***1** to **21***a***5** and the ground conductor layer **23** (described later). Alternatively, the conductor pattern **22** may be indirectly provided to the main surface **211** of the substrate **21**, and the ground conductor layer **23** may be indirectly provided to the main surface **212** of the substrate **21**. For example, another layer having a low conductivity (e.g., a dielectric layer) may be provided (i) between the main surface **211** and the conductor pattern **22** and/or (ii) between the main surface **212** and the ground conductor layer **23**. The substrate **21** includes, in its inside, the later-described conductor posts **21***b***1** to **21***b***5**.

(77) (Conductor Pattern)

(78) Similarly to the conductor pattern **12**, the conductor pattern **22** provided to the main surface **211** can be obtained by patterning of a conductor film into a given shape. The conductor pattern **22** includes strip-shaped conductors **22***a***1** to **22***a***5**, a coplanar line **22***b*, and a coplanar line **22***c*.

(79) The strip-shaped conductors **22***a***1** to **22***a***5** are configured similarly to the strip-shaped conductors **12***a***1** to **12***a***5** in the filter device **1**. Note that each strip-shaped conductor **22***ai* is configured to have a length shorter by a thickness of the substrate **21** than a length of each strip-shaped conductor **12***a***1** in the filter device **1**. The reason for this is that each conductor post **21***bi*

(described later) functions, together with each strip-shaped conductor **22***ai*, as a signal line of a two-conductor line.
(80) The coplanar lines **22***b* and **22***c* are identical to the coplanar lines **12***b* and **12***c* in the filter device **1**. Therefore, the coplanar lines **22***b* and **22***c* will not be described here.
(81) (Recessed Portion)
(82) The recessed portions **21***a***1** to **21***a***5** provided to the main surface **212** are configured similarly to the recessed portions **11***a***1** to **11***a***5** in the filter device **1**. Thus, each recessed portion **21***ai* corresponds to its respective strip-shaped conductor **22***ai* facing the each recessed portion **21***ai*. Note that each recessed portion **21***ai* is configured to have a length shorter than a length of each recessed portion **11***ai* in the filter device **1**. Thus, in the filter device **2**, one end portion of each strip-shaped conductor **22***ai* protrudes from its respective recessed portion **21***ai* overlapping the each strip-shaped conductor **22***ai* in plan view (see (a) and (c) of FIG. **6**).
(83) When seen along a lengthwise direction of each strip-shaped conductor **22***ai* (see (c) of FIG. **6**), a position where the recessed portion **21***ai* is to be provided is defined such that a distance between each conductor post **21***bi* (described later) and a portion of the second ground conductor layer **232** which portion is close to the conductor post **21***bi* is substantially equal to a distance between the strip-shaped conductor **22***ai* and the recessed portion **21***ai*. To be more specific, a position where each recessed portion **21***ai* is to be provided is defined such that a degree of coupling between its respective conductor post **21***bi* and the portion of the second ground conductor layer **232** which portion is close to the respective conductor post **21***bi* (i.e., a portion of the second ground conductor layer **232** which portion covers a side surface of the each recessed portion **21***ai*, the side surface is close to the each conductor post **21***bi*) is substantially equal to a degree of coupling between the respective strip-shaped conductor **22***ai* and a portion of the second ground conductor layer **232** which portion is provided in a bottom surface of the each recessed portion **21***ai*.
(84) In Embodiment 2, each recessed portion **21***ai* has a rectangular parallelepiped shape. However, the shape of each recessed portion **21***ai* can be selected as appropriate, similarly to the shape of each recessed portion **11***ai*. The shape of each recessed portion **21***ai* may be identical to the shape of each recessed portion **11***ai* of the filter device **1**A or to the shape of each recessed portion **11***ai* of the filter device **1**B.
(85) (Ground Conductor Layer)
(86) As shown in (b) and (c) of FIG. **6**, similarly to the ground conductor layer **13**, the ground conductor layer **23** is constituted by a first ground conductor layer **231** and a second ground conductor layer **232**. The first ground conductor layer **231** corresponds to the first ground conductor layer **131** of the ground conductor layer **13**, and the second ground conductor layer **232** corresponds to the second ground conductor layer **132** of the ground conductor layer **13**. The first ground conductor layer **231** refers to a portion of the ground conductor layer **23** which portion is provided to the main surface **212**, whereas the second ground conductor layer **232** refers to a portion of the ground conductor layer **23** which portion covers a surface of each recessed portion **21***ai*.
(87) (Conductor Post)
(88) Similarly to the conductor posts **11***b***1** to **11***b***5** of the filter device **1**, the conductor posts **21***b***1** to **21***b***5** respectively correspond to the strip-shaped conductors **22***a***1** to **22***a***5**. When the main surface **211** is seen in plan view, each conductor post **21***bi* corresponding to its respective strip-shaped conductor **22***ai* is provided in an area where the one end portion of the respective strip-shaped conductor **22***ai* which one end portion protrudes from its respective recessed portion **21***ai* overlaps the later-described first ground conductor layer **231**. Each conductor post **21***bi* short-circuits the one end portion and the first ground conductor layer **231**. Each conductor post **21***bi* has a given degree of coupling with respect to a portion of the second ground conductor layer **232** which portion is close to the each conductor post **21***bi*. Thus, the each conductor post **21***bi*

constitutes a two-conductor line, together with the portion of the second ground conductor layer **232**.
(89) As described above, in the filter device **2**, not only each strip-shaped conductor **22***ai* but also each conductor post **21***bi* functions as a signal line of the two-conductor line. Thus, a length of each strip-shaped conductor **22***ai* can be reduced by a thickness of the substrate **21** than the length of each strip-shaped conductor **12***a***1** in the filter device **1**.
(90) In Embodiment 2, each conductor post **21***bi* is constituted by four conductor posts. However, there is no limitation on the number of conductor posts constituting each conductor post **21***bi*. In order to reduce a difference between a width of each strip-shaped conductor **22***ai* and an effective width of each conductor post **21***bi*, it is preferable to employ the following configuration. That is, (1) in a case where conductor posts constituting each conductor post **21***bi* are separated from each other, the sum of diameters of the conductor posts constituting the conductor post **21***bi* is close to the width of the strip-shaped conductor **22***ai*. Meanwhile, (2) in a case where conductor posts constituting each conductor post **21***bi* are integrated together, the width of the each conductor post **21***bi* (i.e., a length of each conductor post **21***bi* in the width direction of each strip-shaped conductor **22***ai*) is close to the width of the strip-shaped conductor **22***ai*.
(91) <Variations>
(92) The following description will discuss, with reference to FIG. **7**, a filter device **2**A, which is variation of the filter device **2** shown in FIG. **6**. FIG. **7** is an enlarged plan view of one end portion of a strip-shaped conductor **22***a***3**, which is one of strip-shaped conductors included in the filter device **2**A. Note that, for convenience, members of the filter device **2**A having functions identical to those of the respective members described for the filter device **2** are given respective identical reference numerals, and a description of those members is omitted here.
(93) The filter device **2**A can be obtained by modifying the filter device **2** so as to change the shape of each conductor post **21***bi*. FIG. **7** illustrates a conductor post **21***b***3** as an example of each conductor post **21***bi*. The other conductor posts **21***b***1**, **21***b***2**, **21***b***4**, and **21***b***5** have identical configurations to the conductor post **21***b***3**.
(94) Specifically, each conductor post **21***bi* in the filter device **2** is constituted by four conductor posts each having a circular cross-sectional shape. Meanwhile, each conductor post **21***bi* in the filter device **2**A is constituted by eight conductor posts each of which has a circular cross-sectional shapes and adjacent ones of which has a center-to-center distance shorter than a diameter of each conductor post. Thus, when seen along a width direction of each strip-shaped conductor **22***ai*, a width of each conductor post **21***bi* in the filter device **2**A is substantially equal to a width of each strip-shaped conductor **22***ai*.
(95) In Embodiment 2, the width of each conductor post **21***bi* is 92.5% of the width of each strip-shaped conductor **22***ai*. However, the width of each conductor post **21***bi* is not limited to this. In order to improve the degree of continuity between each strip-shaped conductor **22***ai* and each conductor post **21***bi*, the width of each conductor post **21***bi* is preferably in a range of not less than 80% and not more than 120% with respect to the width of each strip-shaped conductor **22***ai*.
EXAMPLES
(96) The following description will discuss, with reference to FIGS. **8** and **9**, a filter device **1**E that is an Example of the present invention and a comparative example of the filter device **1**. FIG. **8** is a plan view of the filter device **1**E. FIG. **9** is a graph showing a result, obtained by simulation, of frequency dependencies of transmission intensities of the filter device **1**E, Comparative Example 1, and Comparative Example 2. Hereinafter, the frequency dependency of the transmission intensity will be referred to as a transmission property.
(97) The filter device **1**E is a variation of the filter device **1**B shown in FIG. **3**. The filter device **1**E was obtained by modifying the filter device **1**B such that the recessed portions **11***ai* was changed from the ones seeming to have an E-shape in plan view to recessed portions **11***ai***1** and **11***ai***2**, which were two independent recessed portions. Each of the recessed portions **11***ai***1** and **11***ai***2** had a

rectangular parallelepiped shape. Note that each recessed portion **11***ai* in the filter device **1**E did not include the fourth recessed portion **11***ai***4** provided in each recessed portion **11***ai* in the filter device **1**B. Note also that a length of each of the recessed portions **11***ai***1** and **11***ai***2** was identical to a length of each strip-shaped conductor **12***ai*.
(98) In this Example, the filter device **1**E employed the following design parameters. Specifically, quartz glass was employed as a dielectric constituting a substrate **11**. A specific inductive capacity thereof was 3.82, and a thickness of the substrate **11** was 400 μm. Each strip-shaped conductor **12***a***1** had a length of 1550 μm and a width of 350 μm. A distance between center axes of adjacent ones of the strip-shaped conductors **12***a***1** was 700 μm. The recessed portions **11***ai***1** and **11***ai***2** constituting each recessed portion **11***ai* each had a length of 1550 μm, a width of 100 μm, and a depth of 250 μm.
(99) A filter device in accordance with Comparative Example 1 was obtained by modifying the filter device **1** so that the recessed portions **11***ai* were excluded therefrom. Thus, in Comparative Example 1, a main surface **112** was made of a flat surface, and a ground conductor layer **13** was made only of a first ground conductor layer **131**. In the filter device in accordance with Comparative Example 1, adjacent ones of strip-shaped conductors were arranged in a manner as illustrated in FIG. 4 of Patent Literature 1. A filter device in accordance with Comparative Example 2 was obtained by modifying the filter device in accordance with Comparative Example 1 such that each recessed portion was provided in an area of a main surface **111** in which no strip-shaped conductor **12***a***1** was provided and which was sandwiched between adjacent ones of the strip-shaped conductors. The filter device in accordance with Comparative Example 2 corresponds to the filter device illustrated in FIG. 1 of Patent Literature 1.
(100) In a configuration in which each of the strip-shaped conductors functions as a resonator and adjacent ones of the strip-shaped conductors are electromagnetically coupled to each other, as in the filter device **1** and the filter devices of Comparative Examples, it is known that a coupling coefficient k between the resonators is expressed by the following formula (1):
(101) $$k = \frac{f_h^2 - f_l^2}{f_h^2 + f_l^2} \quad (1)$$
(102) Here, the coupling coefficient k is an indicator indicating a degree of coupling between the resonators. A greater coupling coefficient k indicates a higher degree of coupling between the resonators. In formula (1), f.sub.h denotes a resonance frequency on a higher frequency side, and f.sub.l denotes a resonance frequency on a lower frequency side.
(103) As shown in FIG. **9**, coupling coefficients k obtained from the transmission properties of the Example, Comparative Example 1, and Comparative Example 2 were 0.0854, 0.184, and 0.149, respectively. This reveals the following. That is, in a case where the filter device **1**E of the Example is designed such that the degree of coupling between adjacent ones of the strip-shaped conductors **12***ai* is substantially equal to those of the filter devices of the Comparative Examples, a distance between adjacent ones of the strip-shaped conductors **12***a***1** can be reduced in the filter device **1**E, as compared to those in the filter devices of Comparative Examples 1 and 2. That is, the above result reveals that the filter device **1**E can be more reduced in size than the filter devices of Comparative Examples 1 and 2.
(104) Aspects of the present invention can also be expressed as follows:
(105) A filter device in accordance with a first aspect of the present invention includes: a substrate which is made of a dielectric and which includes a first main surface and a second main surface facing each other; strip-shaped conductors which are provided to the first main surface and adjacent ones of which are electromagnetically coupled to each other; and a ground conductor layer provided at least to the second main surface, wherein in the second main surface of the substrate, one or more recessed portions are provided for each of the strip-shaped conductors, the one or more recessed portions overlapping the each of the strip-shaped conductors when seen in plan view, the one or more recessed portions having a surface covered with the ground conductor layer.

(106) As compared to a filter device including a substrate without a recessed portion (e.g., the filter device illustrated in FIG. 3 of Patent Literature 1), the above configuration can reduce a distance between adjacent ones of the strip-shaped conductors when the filter device is designed such that a degree of coupling between the adjacent ones of the strip-shaped conductors is substantially equal to those of the conventional ones. Thus, the filter device can be reduced in size. The reason for this is as follows. That is, as compared to the filter device including the substrate without a recessed portion, the above configuration involves a shorter distance between each strip-shaped conductor and a portion of the ground conductor layer which portion is closest to the each strip-shaped conductor, and accordingly lines of electric force generated between the strip-shaped conductors and the ground conductor layer are concentrated in a direction normal to the first main surface and are hardly expanded in an in-plane direction of the first main surface.
(107) A filter device in accordance with a second aspect of the present invention employs, in addition to the feature of the filter device in accordance with the first aspect above, a feature wherein: when seen along a lengthwise direction in which the strip-shaped conductors extend, each of the recessed portions (i) has a length longer than a length of one of the strip-shaped conductors overlapping the recessed portion in the plan view and (ii) covers the one of the strip-shaped conductors.
(108) With the above configuration, the ground conductor layer provided to the bottom surfaces of the recessed portions has a sufficient size as a ground conductor layer constituting a microstrip line.
(109) A filter device in accordance with a third aspect of the present invention employs, in addition to the feature of the filter device in accordance with the first or second aspect above, a feature wherein: the second main surface of the substrate has recessed portions provided for each of the strip-shaped conductors, the recessed portions overlapping the each of the strip-shaped conductors when seen in plan view, the recessed portions having a surface covered with the ground conductor layer.
(110) With the above configuration, it is possible to reduce the volume of each recessed portion provided to the substrate, thereby making it possible to reduce the number of steps, time, and/or the like required to form the recessed portions.
(111) A filter device in accordance with a fourth aspect of the present invention employs, in addition to the feature of the filter device in accordance with any one of the first to third aspects above, a feature wherein: one or more conductor posts are provided, for each of the strip-shaped conductors, in an area where the each of the strip-shaped conductors and a respective one of the recessed portions overlap each other in the plan view, the one or more conductor posts short-circuiting the each of the strip-shaped conductors and the ground conductor layer.
(112) With the above configuration, it is possible to short-circuit the strip-shaped conductors and the recessed portions via the short conductor posts, thereby making it possible to provide a one-end short-circuited strip resonator having a minimum reactance.
(113) A filter device in accordance with a fifth aspect of the present invention employs, in addition to the feature of the filter device in accordance with the fourth aspect above, a feature wherein: the one or more conductor posts are disposed in an area overlapping the respective one of the recessed portions in plan view, and a distance between (i) an area of a bottom surface of the respective one of the recessed portions which area includes the one or more conductor posts and (ii) the first main surface is constant.
(114) With the above configuration, the one or more conductor posts are short-circuited to the ground conductor layer only via the bottom surface of the recessed portion. Therefore, it is possible to simplify the shape(s) of the one or more conductor posts. Furthermore, it is possible to achieve the one or more conductor posts having a constant length.
(115) A filter device in accordance with a sixth aspect of the present invention employs, in addition to the feature of the filter device in accordance with the first aspect above, a feature wherein: a portion of the ground conductor layer which portion is provided to the second main surface is

designated as a first ground conductor layer and a portion of the ground conductor layer which portion covers the surface of the recessed portions is designated as a second ground conductor layer; one end portion of each of the strip-shaped conductors protrudes from one of the recessed portions overlapping the strip-shaped conductor in the plan view; and one or more conductor posts are further provided for each of the strip-shaped conductors, the one or more conductor posts being disposed in an area where the one end portion and the first ground conductor layer overlap each other in the plan view, the one or more conductor posts short-circuiting the one end portion and the first ground conductor layer, the one or more conductor posts constituting a two-conductor line together with a portion of the second ground conductor layer which portion covers a side surface of a corresponding one of the recessed portions.

(116) With the above configuration, in addition to the feature wherein each strip-shaped conductor and the portion of the second ground conductor layer which portion is provided to the bottom surface of it respective recessed portion function as a two-conductor line, the one or more conductor posts and the portion of the second ground conductor layer which portion is provided to the side surface of its respective recessed portion also function as a two-conductor line. Thus, in the filter device in accordance with the sixth aspect, the strip-shaped conductors can be reduced in length in the lengthwise direction. Consequently, the filter device can also be reduced in size in the lengthwise direction.

(117) A filter device in accordance with a seventh aspect of the present invention employs, in addition to the feature of the filter device in accordance with the sixth aspect above, a feature wherein: when each of the strip-shaped conductors is seen along a width direction crossing the direction in which the each of the strip-shaped conductors extends, the one or more conductor posts have a width substantially equal to a width of the each of the strip-shaped conductors.

(118) With the above configuration, it is possible to reduce a degree of discontinuity that can occur at a connection point where the strip-shaped conductor and the conductor post, which function as a signal line of the two-conductor line, are connected to each other. Consequently, it is possible to enhance the functionality of the two-conductor line.

(119) A filter device in accordance with an eighth aspect of the present invention employs, in addition to the feature of the filter device in accordance with any one of the first to eighth aspects above, a feature wherein: the filter device further includes a shield which is made of a metal and which covers the strip-shaped conductors while keeping a distance from the strip-shaped conductors.

(120) With the above configuration, even if a metal object gets closer to the strip-shaped conductors from the first main surface side, the shield can shield the strip-shaped conductors from the metal object. Thus, it is possible to reduce a change in filter characteristics that may otherwise occur in such a case.

Supplementary Notes

(121) The present invention is not limited to the embodiments, but can be altered by a skilled person in the art within the scope of the claims. The present invention also encompasses, in its technical scope, any embodiment derived by combining technical means disclosed in differing embodiments.

REFERENCE SIGNS LIST

(122) **1**, **2**: filter device **11**, **21**: substrate **111**, **211**: main surface (first main surface) **112**, **212**: main surface (second main surface) **11***a***1** to **11***a***5**, **21***a***1** to **21***a***5**: recessed portion **11***b***1** to **11***b***5**, **11***c***1** to **11***c***4**, **21***b***1** to **21***b***5**, **21***c***1** to **21***c***4**: conductor post **12**, **22**: conductor pattern **12***a***1** to **12***a***5**, **22***a***1** to **22***a***5**: strip-shaped conductor **12***b*, **12***c*, **22***b*, **22***c*: coplanar line **12***b***1**, **12***c***1**, **22***b***1**, **22***c***1**: signal line **12***b***2**, **12***b***3**, **12***c***2**, **12***c***3**, **22***b***2**, **22***b***3**, **22***c***2**, **22***c***3**: ground conductor pattern **13**, **23**: ground conductor layer **131**, **231**: first ground conductor layer **132**, **232**: second ground conductor layer

Claims

1. A filter device comprising: a substrate which is made of a dielectric and which includes a first main surface and a second main surface facing each other; strip-shaped conductors which are provided to the first main surface and adjacent ones of which are electromagnetically coupled to each other; and a ground conductor layer provided at least to the second main surface, wherein in the second main surface of the substrate, one or more recessed portions are provided for each of the strip-shaped conductors, the one or more recessed portions overlapping the each of the strip-shaped conductors when seen in plan view, the one or more recessed portions having a surface covered with the ground conductor layer, wherein: when seen along a lengthwise direction in which the strip-shaped conductors extend, each of the recessed portions (i) has a length longer than a length of one of the strip-shaped conductors overlapping the recessed portion in the plan view and (ii) covers the one of the strip-shaped conductors.
2. The filter device as set forth in claim 1, wherein: one or more conductor posts are provided, for each of the strip-shaped conductors, in an area where the each of the strip-shaped conductors and a respective one of the recessed portions overlap each other in the plan view, the one or more conductor posts short-circuiting the each of the strip-shaped conductors and the ground conductor layer.
3. The filter device as set forth in claim 1, further comprising: a shield which is made of a metal and which covers the strip-shaped conductors while keeping a distance from the strip-shaped conductors.
4. A filter device comprising: a substrate which is made of a dielectric and which includes a first main surface and a second main surface facing each other; strip-shaped conductors which are provided to the first main surface and adjacent ones of which are electromagnetically coupled to each other; and a ground conductor layer provided at least to the second main surface, wherein in the second main surface of the substrate, one or more recessed portions are provided for each of the strip-shaped conductors, the one or more recessed portions overlapping the each of the strip-shaped conductors when seen in plan view, the one or more recessed portions having a surface covered with the ground conductor layer, wherein: a portion of the ground conductor layer which portion is provided to the second main surface is designated as a first ground conductor layer and a portion of the ground conductor layer which portion covers the surface of the recessed portions is designated as a second ground conductor layer; one end portion of each of the strip-shaped conductors protrudes from one of the recessed portions overlapping the strip-shaped conductor in the plan view; and one or more conductor posts are further provided for each of the strip-shaped conductors, the one or more conductor posts being disposed in an area where the one end portion and the first ground conductor layer overlap each other in the plan view, the one or more conductor posts short-circuiting the one end portion and the first ground conductor layer, the one or more conductor posts constituting a two-conductor line together with a portion of the second ground conductor layer which portion covers a side surface of a corresponding one of the recessed portions.
5. The filter device as set forth in claim 4, further comprising: a shield which is made of a metal and which covers the strip-shaped conductors while keeping a distance from the strip-shaped conductors.